

OSINT Social Media Monitoring Pipeline: Project Report

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1. Introduction

What is OSINT?

Open Source Intelligence (OSINT) refers to the collection and analysis of publicly available information from digital sources for investigative purposes. Unlike traditional intelligence gathering, OSINT leverages information that anyone can access – social media posts, public databases, websites, and other digital footprints. In today's digital age, OSINT has become crucial for cybersecurity threat detection, brand monitoring, investigative journalism, and law enforcement activities.

Lab Objective:

This project aimed to develop an automated OSINT pipeline that collects, processes, and analyzes data from multiple social media platforms simultaneously. The primary goal was to create a unified system that could:

- Monitor multiple social media platforms in real-time
- Standardize data from different sources into a consistent format
- Perform basic analysis including language filtering and sentiment scoring
- Store results for further investigation and trend analysis

The pipeline serves as a foundation for more advanced OSINT operations, demonstrating how

automated tools can enhance digital investigation capabilities.

2. Methodology

Platforms Integrated

The system integrates data collection from nine major social media platforms:

- Twitter: Public tweets and trends monitoring
- Reddit: Forum discussions and community sentiments
- Facebook: Public page posts and content
- Instagram: Public post collection from profiles
- TikTok: Video metadata and content analysis
- Mastodon: Decentralized social media monitoring
- GitHub: Code repository and developer activity tracking
- Snapchat: Public story and content monitoring

Technical Architecture

The pipeline follows a modular approach with these key components:

- Data Collection Layer: Individual collector modules for each platform
- Processing Layer: Text cleaning, language detection, and sentiment analysis
- Storage Layer: SQLite database for structured data storage
- Configuration System: Environment variables for API key management

Tools and Technologies Used:

- Python 3.8+: Primary programming language
- RapidAPI: Third-party API services for multiple platforms
- Instagrapi: Instagram private API integration
- TextBlob: Natural language processing for sentiment analysis

LangDetect: Language identification and filtering

- SQLite: Lightweight database for data storage
- Requests: HTTP library for API communications

Data Processing Workflow

1. Collection: Parallel data gathering from all integrated platforms
2. Cleaning: URL removal, symbol stripping, text normalization
3. Filtering: English language content selection
4. Analysis: Sentiment scoring (-1.0 to +1.0 polarity)
5. Storage: Structured database persistence

3. Results

System Performance

The pipeline successfully collected data from all integrated platforms, though with varying degrees of completeness and reliability. During testing, the system typically processed 50-100

posts per complete execution cycle.

Data Collection

main.py A x BLACKBOX

```
11 from collectors.squid collector import fetch_squid
12 from collectors.hackernews collector import fetch_hackernews
13 from collectors.vk collector import fetch_vk
14 from collectors.snapchat collector import fetch_snapchat
15 from utils.cleaner import clean_text, filter_english
16 from utils.database import save_to_db
17 from utils.sentiment import add_sentiment
18 from pathlib import Path
19 import sqlite3
20 from tabulate import tabulate
21
22 # Set a reliable database path
23 BASE_DIR = Path(__file__).parent
24 DB_PATH = BASE_DIR / "data" / "osint_data.db"
25
26 def normalize_record(item, platform):
27     """Normalize data into a common schema"""
```

Successfully saved 18 normalized multi-platform records to database
Successfully saved 18 normalized multi-platform records to database

platform	user	timestamp	text	url	sentiment
Reddit	HrmBee	1759682242.0	The Sky Isnt Falling but Starlink Satellites Are Watch your head	https://reddit.com/r/technology/comments/1nyt06w/the_sky_isnt_falling_but_starlink_satellites_are/	Neutral
Reddit	lurker_bee	1759719696.0	Arizona sees spike in demand for VPNs following the introduction of age verification laws	https://reddit.com/r/technology/comments/1nz7ygb/arizona_sees_spike_in_demand_for_vpn_folowing/	Neutral
Reddit	Yansde	1759700376.0	ParkMobile pays 1 each for 2021 data breach that hit 22 million	https://reddit.com/r/technology/comments/1nz0yse/parkmobile_pays_1_each_for_2021_data_breach_that/	Neutral
Reddit	ourlifeintoronto	1759703000.0	Way past its prime how did Amazon get so rubbish	https://reddit.com/r/technology/comments/1nz20c6/way_past_it/	Negative

BLACKBOX Agent Open Website Generate Commit Message Ln 4, Col 59 Spaces: 4 UTF-8 CRLF Python BLACKBOX Autocomplete: ON BLACKBOXAI: Open Chat

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Github	manojkmohan5	2025-10-06T03:11:35Z	SemanticCachingFrameworkforContextAwareConversationsAI A conversational AI cache that reduces similar user queries across multiple chats using contextaware embeddings	https://github.com/manojkmohan5/Semantic-Caching-Framework-for-Context-Aware-Conversations-AI	Neutral
Github	Valtherismyzv7P	2025-08-27T12:23:48Z	SmartAITradingBot Trading Smart AI Bot for Ethereum blockchain	https://github.com/Valtherismyzv7P/SmartAI-Trading-Bot	Positive
Github	sianaoon	2025-10-06T03:59:12Z	Model2webapp Dev web app Gen by Ai Agent Somontar Model20	https://github.com/sianaoon/model2-webapp	Neutral
Github	mdrideout	2025-02-09T17:45:25Z	junjo An AI graph workflow library for building dynamic LLM driven workflows	https://github.com/mdrideout/junjo	Neutral
Github	akansham92	2025-10-06T03:56:52Z	UrbanVitalityIndexUVIAnASAEarthObservationPoweredSmartGrowthFramework The Urban Vitality Index UVI is an AIdriven decisionsupport system that integrates NASA Earth observation EO data to help city planners design climateresilient equitable and environmentally balanced urban growth strategies	https://github.com/akansham92/Urban-Vitality-Index-UVI-A-NASA-Earth-Observation-Powered-Smart-Growth-Framework	Neutral
Github	SteelQ	2025-10-06T03:59:16Z	aimvpphgithub	https://github.com/SteelQ/aimvpph-github	Neutral

